

AKROLOY®

PA ICF 30 black (5269)

PA66+PA6I/6T CF30

AKROLOY® PA ICF 30 black (5269) is a 30% carbon fibre reinforced, semi-aromatic polyamide blend with high stiffness, good sliding properties and reduced moisture.

Features

low warpage reduced moisture

Properties

Modulus	Strength	Impact
25.000 MPa	220 MPa	50 kJ/m ²

Sustainability

Recycled content 30 %

Mechanical Properties

Tensile modulus	1 mm/min d.a.m.	25000 MPa
ISO 527-2	1 mm/min conditioned	23000 MPa
Tensile stress at break	5 mm/min d.a.m.	220 MPa
ISO 527-2	5 mm/min conditioned	200 MPa
Tensile strain at break	5 mm/min d.a.m.	1,5 %
ISO 527-2	5 mm/min conditioned	1,6 %
Flexural modulus	2 mm/min d.a.m.	21000 MPa
ISO 178	2 mm/min conditioned	21000 MPa
Flexural strength	2 mm/min d.a.m.	320 MPa
ISO 178	2 mm/min conditioned	300 MPa
Flexural strain at break	2 mm/min d.a.m.	2 %
ISO 178	2 mm/min conditioned	2 %

Disclaimer

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Charpy impact strength ISO 179-1/1eU	23°C d.a.m.	50 kJ/m ²
	23°C conditioned	50 kJ/m ²
	-30°C d.a.m.	40 kJ/m ²
	-30°C conditioned	40 kJ/m ²
Charpy notched impact strength ISO 179-1/1eA	23°C d.a.m.	7 kJ/m ²
	23°C conditioned	7 kJ/m ²
	-30°C d.a.m.	6 kJ/m ²
	-30°C conditioned	6 kJ/m ²

Thermal Properties

Temperature of deflection under load HDT/A ISO 75	1,8 MPa	230 °C
Melting temperature ISO 11357-3	DSC, 10K/min	255 °C

Flammability

Burning rate UL 94	0,8 mm Wall thickness	HB Class
Burning rate (<100 mm/min) FMVSS 302	> 1 mm Thickness	+

General properties

Density ISO 1183	23°C	1,29 g/cm ³
Humidity absorption ISO 1110	70°C, 62% r.H.	1,6 %
Molding shrinkage ISO 294-4	flow	0,1 - 0,3 %
	transverse	0,3 - 0,5 %

Electrical Properties

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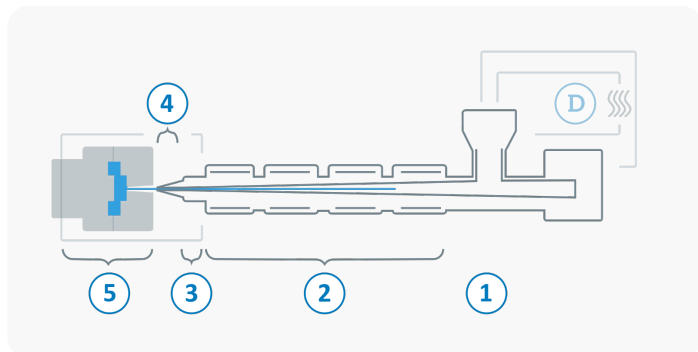
Surface resistivity	d.a.m.	10⁴ Ω
IEC 62631-3-2	conditioned	10⁴ Ω

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Processing

The values mentioned are recommendations. We only recommend desiccant / dry air dryers or vacuum dryers. Too long a drying time and the resulting residual moisture content below the lower limit can lead to filling problems and surface defects. The specified drying time refers to closed and undamaged bagged material. When processing from previously opened bags or from octabins with polyolefin inliners, a longer drying time may be necessary. It is recommended to check the residual moisture content after the drying process.

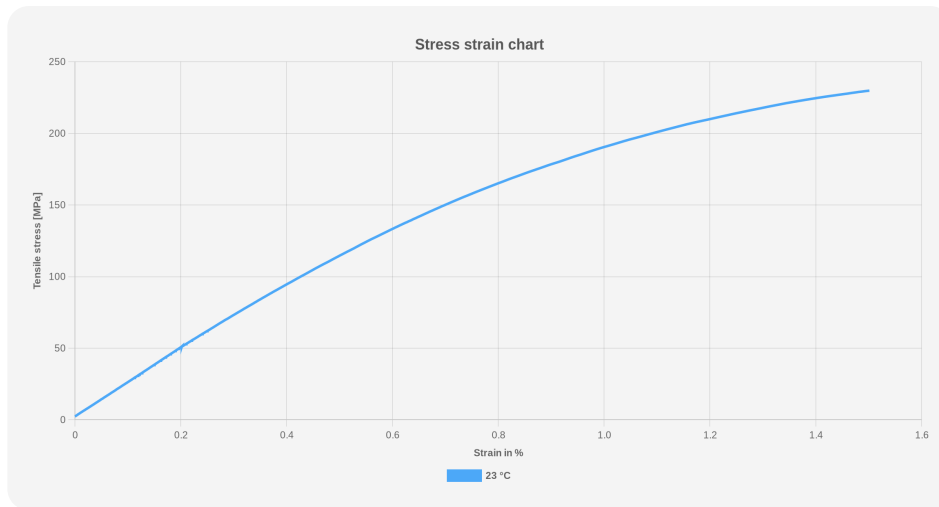


D	Drying time	0 - 4 h
	Drying temperature ($\tau \leq -30^{\circ}\text{C}$)	80 °C
	Processing moisture	0,02 - 0,1 %
1	Feed section	60 - 80 °C
2	Temperature Zone 1 - Zone 4	260 - 310 °C
3	Nozzle temperature	270 - 310 °C
4	Melt temperature	280 - 310 °C
5	Mold temperature	90 - 130 °C
→	Holding pressure, spec.	300 - 800 bar
←	Back pressure, spec.	50 - 150 bar
	Injection speed	medium to high
	Screw speed	8 - 15 m/min

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Diagrams



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